Chapter 1

Raising alternatives to express dependence: The compositional issues

Dependence utterances can be built with an adverbial adjunct phrase, e.g. ”Depending on how the first round goes, I might stop”. They express a complex conditional dependence between two sets of alternatives. Contrary to finite ”depend”, adjunct dependence utterances (ADUs) exhibit diverse syntactic forms and pragmatic purposes that challenge previous approaches. We show how Dynamic Inquisitive Semantics and Free Association with focus solve the compositional issues in English and French. Different dependence types (speech-act level, clause level, and predicate level) are explained by different adjunct attachment sites, with a single uniform entry for ”depend” that captures the various alternative triggers.

1 Introduction

1.1 Main problem

The verb *depend* can express a dependence relation between two questions. For example, sentence (1) says that if it is sunny, Amy will go to the beach. Otherwise, she will not.

(1) [$_B$ Whether Amy will go to the beach] depends on [$_A$ whether it is sunny]. (Hui 2021: 1.3c)

In English, we can also express dependence using the gerundive *depending on* with its complement A and letting this verbal phrase (VP) modify B as a main clause. We call this construction an adjunct dependence utterance (ADU). For example, sentence (2) implies that there are different possible methods to connect two tiles, and that the method used determines whether the tiles are rough on the fingertips.

(2) but [depending on [$_A$ the method you used to connect two tiles]], [$_B$ they could be a little rough on the fingertips] [1]

According to Ciardelli (2018), B depends on A if the resolution of the question A determines the resolution of the question B.[2] For main-clause dependence propositions like (1), this definition works well because A and B are interrogative clauses. However, Ciardelli's characterization fails to capture the intuition that example (2) also expresses dependence, even if the phrases A and B are not congruent to a question: A is a noun phrase (NP) and B is a declarative clause. This constitutes a definition issue. Moreover, this apparent syntactic diversity raises a problem for a unified semantic entry for the verb *depend*. Therefore, we face a compositional issue.

The goals of this chapter are: (i) to describe the diversity of adjunct dependence utterances, (ii) to provide a general definition of dependence, and (iii) to start modeling this diversity uniformly with a single lexical entry for *depend* for clause-level ADUs.

We provide examples selected by hand from the TenTen web corpus family (Jakubíček et al. 2013) and `CIENSFO`, a collection of non-standard French interrogatives (Richard 2024a).

1.2 Outline of this chapter

Section 2 provides novel data revealing the wide range of adjunct dependence utterances, in different languages and with different speech acts, and diagnoses compositional issues. In §3, we present previous attempts to model dependence propositions and show their limits. In §4, §5, and §6, we introduce concepts that help us make sense of the apparent diversity regarding ADU's antecedents, consequents, and semantic type. A general definition of dependence is given in §7. Section 8 explains how different ADU types can be produced by attaching the adjunct on a precise site in the derivation tree. We also propose a uniform lexical entry for *depend* for main-clause ADUs in dynamic inquisitive semantics. Finally, §9 concludes.

[1] The Gallant Goblin. 2020. WarLock Tiles: Stairs & Ladders and EZ Clips Review - WizKids Prepainted Miniature Terrain. YouTube (https://youtu.be/lN-p0YoNWrc) (Accessed 2026-04-13, 0:41–0:52)

[2] To simplify matters, we focus on complete dependence here, excluding cases where a question's resolution might partially depend on another question's resolution. For a discussion, see Karttunen (1977: fn. 6) or Kaufmann (2016: fn. 46).

2 The diversity of adjunct dependence utterances

2.1 Cross-linguistic variation

In French, we can also build adjunct constructions expressing dependence, for example (3). These so-called interrogative-based adverbial adjuncts can be introduced by the (complex) prepositions *selon* 'in accordance with/according to', *suivant* 'following' or *en fonction de* 'in function of' (de Cornulier 2013, Richard 2024a).[3]

(3) des jeunes des jeûnes [$_B$ ça peut être paronyme ou homonyme]
some youngs some fasts it can be paronymous or homonymous
[suivant [$_A$ comment vous le prononcez]]
following how you it pronounce
'[$_B$ French words *jeune* and *jeûne* can be paronymous or homonymous] [depending on [$_A$ how you pronounce *jeûne*]].' [CIENSFO: 050]

Sentence (3) can be translated by an English sentence using the construction *B, depending on A*, similarly to sentence (2). The order between *B* and *depending on A* is not relevant here. This kind of adjunct construction with the same meaning is possible in many European languages, including German (4). Therefore, we will abstract from idiosyncrasies and consider a general form for our object of study.

(4) [Je nachdem, [$_A$ wie alt dein Kind ist]], [$_B$ kann es in allen Bussen
each after.PRO how old your child is can they in all buses
und Bahnen der Leipziger Verkehrsbetriebe gratis mitfahren oder
and trains the.GEN Leipziger Verkehrsbetriebe free travel or
zum ermäßigten Sonder-preis.]
to_the.DAT reduced special-price
'Depending on how old your child is, they can travel free of charge or at a reduced special price on all buses and trains operated by Leipziger Verkehrsbetriebe.' [deTenTen23]

[3] Note that there exists another use of *selon* and *suivant* that can be translated by 'according to', reporting the source of some information, e.g (i). This reading is impossible with *en fonction de*. Thus, we assume this use of *selon/suivant* is independent and we do not focus on it here.

(i) Selon les prévisions météo, le mercure va baisser à partir de mardi.
according_to the forecasts weather the mercury goes drop to leave from Tuesday
'According to the weather forecast, the mercury will drop from Tuesday.' [frTenTen23]

2.2 Adjunct dependence utterances as a general construction

We call *adjunct dependence utterance* (ADU) a clause of any syntactic type that contains a main clause *B* with an XP adjunct made up of an item of category X and its complement *A*, and expressing a dependence relation between the interpretations of *A* and *B*. The adjunct can be before or after *B*. For convenience, we will call and gloss SELON any such adjunct head of category X. In English X = V (verb), in French X = P (preposition), and in German X = C (complementizer). The syntactic structure of an ADU is displayed in Figure 1. By analogy with conditionals used in the paraphrase for finite *depend* proposition (1), we call *antecedent* the phrase *A* and *consequent* the phrase *B*.

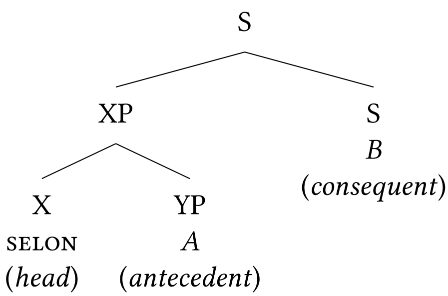

Figure 1: Basic syntactic structure of an adjunct dependence utterance.

We focus on English and French examples in the rest of the article.

2.3 Syntactic and pragmatic diversity

Regardless of the nature of the adjunct clause XP, we observe variation in the syntactic categories that the antecedent *A*, the consequent *B*, and the whole utterance can take.

(5) Selon [$_A$ qu' il s'agit d' un permis A2, A1 ou A], le coût
SELON that it REFL=is_about of a license A2, A1 or A, the cost
n'est pas le même.
NEG=is not the same.
'Depending on whether it's an A2, A1 or A license, the cost is not the same.' [`frTenTen23`]

The antecedent *A* can be an interrogative (3), but it can also be an NP (2) or a declarative complementizer phrase, like in (5). This last construction is only available in French. In all adjunct dependence utterances presented above, the consequent is a declarative clause. Therefore, ADUs accept arguments that are not questions or similar.

In addition, we observe that adjunct constructions expressing dependence need not be assertions. The entire utterance can be a question, as in (6), asking what the egg consumption recommendations are for appropriate age ranges. It can also be a command, like (7), prompting the user to perform one action from a list, and the appropriate action is determined by the user's computer settings. Even if these examples do not assert a state of dependence, they clearly intuitively involve "something depending on something else". Therefore, the definition of dependence should be broadened to include sentences like (6) and (7). We call assertive (resp. inquisitive, imperative) ADUs the adjunct dependence utterances that are assertions (resp. questions, commands).

(6) Selon [$_A$ les âges], [$_B$ combien d' œufs peut-on consommer] ?
SELON the ages how_many of eggs can=we consume
'[$_B$ How many eggs can people eat], depending on [$_A$ their age]?'
[frTenTen23]

(7) [$_B$ Cliquez ou double-cliquez], selon [$_A$ vos paramètres personnels].
click or double-click SELON your parameters personal
'[$_B$ Click or double-click], depending on [$_A$ your personal settings].'
[frTenTen23]

2.4 Diagnosing the problems

Ciardelli's definition of a dependence relation assumes that (i) the antecedent and the consequent are questions and (ii) a dependence relation is a statement. The data provided contradicts these assumptions by showing that (i) the antecedent and the consequent can be syntactically and semantically non-congruent to a question and (ii) a dependence relation can also be a question or a command.

Facing this bewildering diversity, there are two possible solutions. Either Ciardelli's characterization is just inadequate for our data. Or there is a way to coerce or extract content from the respective components to fit his definition.

For ADU's arguments, we will argue that we can retrieve a question-like semantic contribution from diverse antecedents and consequents. More precisely, we will demonstrate that antecedents can be interpreted as inquisitive issues and consequents can raise focus alternatives. Regarding ADU's illocutionary act, we will need to revise the definition of dependence by getting rid of existential closure and truth conditions.

3 A first model for dependence propositions

3.1 Simple examples

In this section, we review a formal model of dependence propositions and show its limitations. Let us bring back example (3), repeated and translated in (8), to provide the complete specification of a dependence proposition on a simple example. In French, there are two ways to pronounce the word *jeûne*, either A_1 = /ʒøn/ or A_2 = /ʒœn/. The word *jeune* is always pronounced /ʒœn/. The consequent B expresses that the words *jeune* and *jeûne* are in a relation, either B_1 = paronymy (almost the same pronunciation) or B_2 = homonymy (the same pronunciation). Richard (2024a) therefore suggests that the meaning of sentence (8) is the conjunction of conditionals: if A_1 then B_1 and if A_2, then B_2.

(8) [$_B$ French words *jeune* and *jeûne* can be paronymous or homonymous] [depending on [$_A$ how you pronounce *jeûne*]].

This simple example reveals three key ingredients in the meaning of dependence: (i) the antecedent and the consequent raise alternatives, (ii) there exists a correspondence between the antecedent's alternatives and the consequent's ones, and (iii) the sentence is interpreted as a conjunction of conditionals. Technically speaking, (8) does not specify the exact correspondence between the pronunciation of *jeûne* and the relation between words *jeune* and *jeûne*. This association is recovered from world knowledge.

Based on Karttunen (1977), Kaufmann (2016) proposes a formal model for *depend(ing on)* for consequents being a disjunction of declaratives or imperatives. Theiler et al. (2019) present a similar model couched in inquisitive semantics. Their formula explicitly mentions a correspondence between alternatives, what will turn out to be crucial in our general definition of dependence. Therefore, we only present Theiler et al.'s semantic framework.

3.2 Background on inquisitive semantics

Inquisitive semantics (Ciardelli et al. 2018) is a formal semantics that treats declaratives and interrogatives in an integrated way. Both syntactic clause types are interpreted by the same semantic object: an *issue*. Issues can have semantic properties independently of the type of the clause they come from.

An issue is a downward-closed set of classical propositions and has the type $\langle\langle e, t\rangle, t\rangle$. An issue is evaluated against a classical proposition s instead of a single world $w \in s$. Issues have alternatives. The set of alternatives $\textsc{alt}(I)$ of an issue I is the set of the least informative propositions $s \in I$ (i.e. the maximal elements for set inclusion). For example, the alternatives of issue $\{\{w_1, w_2\}, \{w_1\}, \{w_2\}, \{w_3\}, \emptyset\}$ are the sets $\{w_1, w_2\}$ and $\{w_3\}$.

Intuitively, the alternatives of an issue are the most general propositions that the issue entails. If I has only one alternative, it supports only one proposition, and so it is called assertive. If I has more than one alternative, it is open to several propositions and is therefore called inquisitive. The alternatives of inquisitive issues are the possible answers to the questions they represent.

(9) Tu veux du thé ou du café ?
you want some tea or some coffee
'Do you want tea or coffee?'

A key feature of the compositional theory of inquisitive semantics (Ciardelli et al. 2017) is that disjunction creates alternatives by default. This allows declarative clauses with a disjunction to be interpreted as inquisitive issues (supposing the intonation supports a questioning speech act). For example, (9) is syntactically declarative according to the Grande Grammaire du Français (Delaveau et al. 2021), but is interpreted as an inquisitive issue whose alternatives are "you want tea" and "you want coffee".

3.3 Theiler et al.'s model

Theiler et al. 2019 observe that in sentence (10), the phrase M indicates that the dependence between the income tax rate and the age only applies to Dutch citizens. They interpret M as a modal base and include it in the definition of dependence as a restriction on the conjunction of conditionals. In most cases, this modal base is left implicit.

(10) $[_M$ According to Dutch law$]$, $[_B$ one's income tax rate$]$ depends on $[_A$ one's age$]$. (Theiler et al. 2019: 39)

Let us write $\mathsf{dep}_M(A, B)$ the interpretation of an adjunct dependence utterance SELON A, B or B, SELON A. Theiler et al.'s model is given in (11) and (12). We slightly modified their formula to make sure the output $\mathsf{dep}_M(A, B)$ is also an issue.

(11) $\mathsf{dep}_M(A, B) := \lambda s.\ \forall w \in s.\ \exists f : \textsc{alt}(A) \to \textsc{alt}(B).\ \mathsf{fmd}(f, M)$

(12) $\mathsf{fmd}(f, M) = (12a) \wedge (12b)$, where

a. **conjunction of conditionals:**
$\forall w \in M.\ \forall t \in \textsc{alt}(A).\ w \in t \rightarrow w \in f(t)$, and

b. **non-triviality:**
$\exists t, t' \in \textsc{alt}(A).\ t \cap M \neq \varnothing \wedge t' \cap M \neq \varnothing \wedge f(t) \neq f(t')$

Definition (11) declares that a dependence proposition is interpreted as an assertive issue containing a single alternative. In all worlds w of that alternative, there exists a function associating a consequent alternative to every antecedent alternative, and such that f is a functional modal dependence (fmd) with respect to a given modal base.[4] Functional modal dependence is the conjunction of two conditions. The first one is a conjunction of the following conditional statements: if the antecedent alternative t is true, then the consequent alternative $f(t)$ is also true. This expression is similar to the example with *jeûne*, but restricted to worlds belonging to the modal base M. The second condition (12b) eliminates trivial dependence (i.e. constant functions f) by requiring that at least some inputs t, t' intersecting with M are mapped to different outputs $f(t) \neq f(t')$.

Non-triviality condition (12b) requires semantic arguments A and B to be inquisitive. Any non-disjunctive declarative as antecedent or consequent of dep would be interpreted as an assertive issue and would thus lead to a contradiction because of this non-triviality. Theiler et al. (2019) argue that this systematic contradictoriness explains the ungrammaticality of (non-disjunctive) declaratives as subject or object of *depend*. This analysis correctly predicts the judgments in (13).

(13) a. * That the light is on depends on whether the switch is up.

b. * Whether the light is on depends on that the switch is up.

Although this model works well for finite *depend*, it faces the same problems as Ciardelli 2018. It cannot account for ADUs where the consequent is a non-disjunctive declarative, e.g. (2). Moreover, it cannot model inquisitive and imperative ADUs. Therefore, we need a broader definition of dependence. Sections 4–6 investigate the variation presented in §2 and propose solutions to extend previous definitions of dependence to capture our data.

4 The antecedent

The antecedent of an adjunct dependence utterance can be an interrogative clause (3), an NP (2), or a declarative complementizer phrase (5) (in French only). De-

[4] In practice, the modal base should vary with w, but here we keep it constant for simplicity. See Richard (2026: §6.5) for a version with a modal base function.

spite the syntactic mismatch, we can interpret all of these phrases as inquisitive issues.

Interrogatives are interpreted as inquisitive issues (Ciardelli et al. 2017). In particular, wh-words raise as many alternatives as individuals in their domains. For example, interrogative (14a) raises the alternative set (14b).

(14) a. how old your child is [*A* in (4)]

b. {“your child is 1 y.o.”, “your child is 2 y.o.”, “your child is 3 y.o.”,…}

Some noun phrases can be interpreted as concealed questions (Baker 1968, Grimshaw 1979, Frana 2017, among many others). For example, the NP (15a) can be coerced to the identity question (15b), whose alternative set is (15c). Concealed-question reading is common with relational nouns or relativized DP, like in (2). We can use the concealed question model of Aloni & Roelofsen (2011) and Hui (2021) in inquisitive semantics to interpret NPs as inquisitive issues.

(15) a. the ages [*A* in (6)]

b. what the age is

c. {“the age is 1 y.o.”, “the age is 2 y.o.”, “the age is 3 y.o.”,…}

Declarative complementizer phrases are interpreted as inquisitive issues when they include a high-scoping disjunction. A declarative complementizer phrase with no explicit disjunction cannot be the complement of a French ADU (16). The disjunction in example (5) scopes over the complementizer phrase (CP) and can be rephrased by repeating a whole CP, like in (17).

(16) *Selon qu’ il s’agit d’ un permis A, le coût n’est pas le même.
SELON that it REFL=is_about of a license A the cost NEG=is not the same
(‘Depending on {whether it’s an A license / which license it is}, the cost is not the same.’)

(17) Cet effort n’est pas le même selon [$_A$ [$_{CP}$ qu’ on est héritier] ou [$_{CP}$ que l’on a que sa force de travail]].
this effort NEG=is not the same SELON that one is heir or that one has only one’s force of work
‘This effort is not the same for heirs as it is for those who only have their labor power.’ [`frTenTen23`]

This kind of CP disjunction can also occur in French concessive (18a) and in the complement of (ir)relevance verbs (18b).

(18) a. Que vous accueilliez un nouveau salarié ou que l' un d'
that you welcome.SJV a new employee or that the one of
entre eux doive vous quitter, nous sommes à vos côtés.
between them must.SJV you leave, we are at your sides.
'Whether you're welcoming a new employee or one of your staff is leaving, we're here to help.' [frTenTen23]

b. La loi se rappellera à vous, car elle se fout
the law REFL call.FUT to you because she REFL not_give_a_damn
que vous ayez été payé ou non.
that you have.SJV been paid or not
'The law will remember you, because it doesn't care whether you've been paid or not.' [frTenTen23]

Inquisitive-issuing CP disjunction is impossible in English (19a), *whether* must be used instead. Roelofsen & Dotlačil (2023) argue that disjunction scoping under the (overt or covert) declarative complementizer does not raise alternatives. This property explains why any complementizer phrase is forbidden in antecedents of English ADUs, even with disjunction (19b).

(19) a. * This is not the same depending on that you inherited or that you must work.

b. * Depending on that it's a A1, A2 or A license, the cost is not the same.

We showed that, despite syntactic variation in ADU antecedents, we can always interpret them as inquisitive issues, thereby saving Theiler et al.'s model.

5 The consequent

5.1 Compositional issue in the consequent

In §5, we focus on assertive adjunct dependence utterances, i.e. the ones for which the whole utterance is an assertion. These ADUs have in common that their consequent B is a declarative clause. So how can they be arguments of a dependence relation? In some cases, like (4) and (8), the consequent B contains a disjunction. This feature could be used to interpret B as an inquisitive issue. However, some consequents contain no disjunction.

Despite this apparent paradox, it is intuitively possible to retrieve alternatives from declarative consequents without disjunction. Contrary to what Kaufmann (2016: §4.3) suggests, various kinds of alternatives are available: domain alternatives of an indefinite adjective (e.g. *certain* in 20), lexical alternatives (e.g. *stop* in 21), and scalar alternatives (22). The alternative triggers are marked in small caps in the original text and the English translation.

(20) on projette le fait que selon comment une personne parle [$_B$ elle
one projects the fact that SELON how a person speaks she
aurait une CERTAINE identité sociale]
has.COND a certain identity social
'We project the idea that, depending on how someone speaks, [$_B$ they would have a CERTAIN social identity].' [CIENSFO: 308]
   a. e.g. B_1 = "they would be a professor", B_2 = "they would be a laborer", B_3 = "they would be an aristocrat",...

(21) suivant comment se passe le premier tour [$_B$ je vais peut-être
SELON how REFL goes the first round I go maybe
ARRÊTER]
stop
'Depending on how the first round goes, [$_B$ I might STOP].' [CIENSFO: 161]
   a. B_1 = "I stop", B_2 = "I continue"

(22) le sexisme suivant quel type de sexisme [$_B$ y a eu des
the sexism SELON which type of sexism there has been some
associations qu' étaient PAS TRÈS FORTES]
associations that were not very strong
'Depending on what type of sexism, [$_B$ there were associations (between sexism and hatred of inclusive writing) that were NOT VERY STRONG].' [CIENSFO: 324]
   a. B_1 = "there were strong associations", B_2 = "there were not very strong associations"

Some cases do not seem to contain an explicit trigger, e.g. (2), (23). By default, they are interpreted as polar: the surface alternative B_1 and its negation B_2.

(23) Suivant où se trouve ta fuite [$_B$ pas la peine de tout
SELON where REFL find your leak not the trouble of all
vidanger].
drain
'Depending on where your leak is, [$_B$ there's no need to drain

everything].' [frTenTen23]

a. B_1 = "no need to drain everything", B_2 = "everything must be drained"

To understand what's going on here, we must first introduce free association with focus.

5.2 Background on free association with focus

Quantificational adverbs (e.g. *always*) quantify over the situations described in a clause. Like generalized quantifiers, they have a restrictor and a nuclear scope. For example, $\mathsf{always}(s, \mathsf{rain})$ indicates that, in all worlds $w \in s$, it is raining. The restrictor of quantificational adverbs is also called its modal base.

Quantificational adverbs are sensitive to focus. They have different meanings depending on which item is focused on in the sentence. For instance, (24) with the stress on NICE means something slightly different from (25), with the stress on SANDY.

(24) Kim always tells Sandy to be NICE. (and not something else)

a. e.g. B_1 = Kim tells Sandy to be nice, B_2 = Kim tells Sandy to be strong,...

(25) Kim always tells SANDY to nice. (and not someone else)

a. e.g. B_1 = Kim tells Sandy to be nice, B_2 = Kim tells Mary to be nice,...

According to Rooth 1985, focus raises alternatives. Free association with focus (Beaver & Clark 2008) describes how the implicit modal base of quantificational operators is provided as the union of these alternatives. For example, the reading (24) means that, in all cases where Kim tells Sandy something, that thing is "being nice". The alternatives are (24a), and their union composes the modal base $M = \bigcup_i B_i$. Sentence (24) is represented by formula $\mathsf{always}(M, B_1)$, paraphrased as "In all worlds $w \in M$, B_1 is true". Conversely, the modal base of (25) consists of the situations where Kim tells somebody to be nice.

5.3 SELON is focus-sensitive

Adjunct dependence utterances are sensitive to stress in their consequents. Compare (26) with a stress on COOK and (27) with a stress on MARIE. In (26), we infer that if Marie is not asked to cook, she will be asked to do something else. For example, reading (26) is felicitous in a context where Marie is assigned to an unimportant task, but if time runs out, she will be reassigned to a more urgent task, like cooking. Conversely, (27) implies that if Marie is not asked to cook, someone else will be. This reading is felicitous in a context where someone is already assigned to cooking, but if time runs out, Marie will cook instead of that person, for example, because she cooks faster. With a sentential focus, as in (28), a polar interpretation is derived by default.

(26) Depending on the time, I might ask Marie to COOK. (and not something else)

a. B_1 = "I ask Marie to cook", B_2 = "I ask Marie to do the laundry",...

(27) Depending on the time, I might ask MARIE to cook. (and not someone else)

a. B_1 = "I ask Marie to cook", B_2 = "I ask Jean to cook",...

(28) Depending on the time, I MIGHT ASK MARIE TO COOK.

a. B_1 = "I ask Marie to cook", B_2 = "I don't ask Marie to cook"

The idea that *depend* is linked to focus has been proposed before. Kaufmann (2016) proposes that, in the constructions "*Depending on Q, p*", p raises "live issues". Hénot-Mortier 2024b assimilates these "live issues" with the maximal true answers to the question under discussion (QuD) evoked by p. This question under discussion can be interpreted as a question inquiring about p's focused material (Hénot-Mortier 2024a).

Based on these pieces of evidence, the adjunct SELON A qualifies as a complex quantificational operator and free association with focus applies. Focus in the consequent of an ADU raises focus alternatives B_i. The union of these focus alternatives determines a modal base. By default, the modal base of $\mathtt{dep}_M(A, B)$ in formula (11) is this derived modal base $M = \bigcup_i B_i$.

Consequent alternatives are triggered by focus. However, Theiler et al.'s definition is not entirely saved because focus alternatives are not at-issue and consequents are still interpreted as assertive issues. Section 8.2 solves this problem by proposing a uniform derivation of inquisitive and focus alternatives.

5.4 Modal base quantification

A recurring feature of assertive ADUs is the presence of a possibility modal in the consequent, e.g. (2) and (21), repeated and translated in (29). Sentence (29) may be paraphrased as (29a), including *might* in the consequent alternative. However, there is another reading excluding *might* in the consequent alternative, but with an implicature about the overall possibility of the surface alternative.

(29) Depending on how the first round goes, I might stop.

a. If the round goes badly, I might strop, and if it goes well, I might continue.

b. If the round goes badly, I stop, and if it goes well, I continue. And it might be that (the round goes badly and) I stop.

In that second reading, the operator *might* is anchored to the whole utterance. Such an epistemic modality is necessary to suggest alternatives in consequent's default polar interpretation because it shows that the speaker considers both the surface consequent and its negation as possible. Moreover, its modal base is given by the ADU's modal base, thus providing the additional inference $\mathsf{might}(M, B_1)$.

5.5 Predicate-level dependence adjuncts

The adjunct dependence utterances where the main-clause predicate expresses variation (*is not the same*, *is different*, *varies*,...) or selection (*choose*,...) seem to behave differently. These predicates quantify on the alternatives raised by their subjects, interpreted as concealed questions. Take example (5) repeated and translated in (30). This sentence cannot be paraphrased by keeping predicate *is (not) the same* in the conditionals (30a). The expected reading is a conjunction of the meaning of the predicate and dependence (30b). Moreover, only *the cost* is the semantic consequent of *depending on*, example alternatives being (30c).

(30) Depending on whether it's an A2, A1 or A license, [$_B$ the cost] is not the same.

a. ≠ If it's an A2 license, the cost the not the same, if it's an A1 license, the cost is the same,...

b. The cost (of a license) varies and it depends on whether it's an A2, A1 or 1 license.

c. B_1 = "the cost is €700", B_2 = "the cost is €900",...

Sentence (31) has a similar pattern. TI cannot be rephrased by something akin to (31a) (without neg-raising). It rather contributes a conjunction between a choice decision and the specification of a selection criterion (31b).

(31) je choisis [$_B$ mes activités] en fonction de si y a un train qui y va
I choose my activities SELON whether there has a train that there goes
'I choose [$_B$ my activities] based on whether there is a train that goes there.' [`CIENSFO: 008`]

a. $\neq$ If there's a train going to activity a, then I choose to go to activity a, if there's no train going to activity b, then it's false that I choose to go to activity b,...

b. I choose my activities and the criterion is a (non-trivial) function from activities to whether there's a train going there.

Richard (2024a: §3.2.6) observes that the SELON A phrase in (31) exhibits a stronger syntactic connection with the main clause. However, even in (31), this phrase is syntactically and semantically optional. Therefore, we consider predicate-level constructions with SELON as regular adjuncts.

We call *clause-level ADUs* the adjunct dependence utterances where SELON A semantically modifies the whole clause it syntactically modifies. The examples seen in §5.5 semantically modify the predicate meaning and not the whole utterance. In particular, the consequent is given by the predicate's subject and not the whole main clause. Therefore, we refer to them as *predicate-level ADUs*.

6 Non-assertive adjunct dependence utterances

6.1 Imperative adjunct dependence utterances

Examples (7) and (32) show that adjunct dependence utterances can have a directive illocutionary act. In these cases, the consequent is an imperative clause. The consequent alternatives are commands (in a broad sense), e.g. (32a). Instruction (32) can be paraphrased as a conjunction of conditionalized commands, as in (32b).

(32) [$_B$ Buy some teak or mahogany], depending on which is in stock. (Fox 2015: 17b)

a. B_1 = "buy some teak", B_2 = "buy some mahogany"

b. If there is teak in stock, buy some teak, if there is mahogany in stock, buy some mahogany.

Theiler et al. (2019) cannot model imperative ADUs because commands do not have truth conditions but rather satisfaction conditions (Fox 2015). For example, "buy some teak" is satisfied if the hearer x commits to make the proposition "x bought some teak" true. Therefore, a definition of dependence must abstract from truth conditions to include imperative ADUs.

Imperative ADUs often come with an exclusivity inference: the antecedent alternatives have an empty intersection, and so, at most one of the consequent commands should be satisfied. This inference is an optional implicature derived from the disjunction in the imperative consequent, and is independent of the ADU. In (32), this exclusivity inference is forced by singular *which*, indicating that only one antecedent alternative holds. However, with a number-unmarked wh-word, this implicature can be waived, as exemplified in (33).

(33) To secure your account, please activate two-factor authentication or biometrics, depending on what suits you. Use both methods to ensure maximum security.

6.2 Inquisitive adjunct dependence utterances

6.2.1 Speech-act-level dependence

Adjunct dependence utterances can be questions. In this case, the consequent is an interrogative clause. For example, question (34) is equivalent to a conjunction of separate questions (34a). Similarly to a universally quantified reading, all of these questions need to be answered. Like for imperative ADUs, the dependence relation scopes over the whole questioning speech act, so we call (34) and (32) *speech-act-level ADUs*.

(34) What do we do about the forest fires (hurricanes, flooding, heat waves, depending on where you live)? [`enTenTen21`]
   a. If we live in a place prone to forest fires, what do we do about forest fires? If we live in a place prone to hurricanes, what do we do about hurricanes?...

6.2.2 Questions about dependence

Contrary to question (34), most inquisitive ADUs do not raise multiple questions but a single one. For example, sentence (6), repeated and translated in (35), requests age ranges and a dependence function from these age ranges to consumption recommendation regarding eggs. The answer must be a list of pairs describing that function, like (35a).[5] The second question of (36) wonders whether there exists a non-trivial dependence function from moon phases to grass growth speed. We call this kind of inquisitive ADUs *dependence questions*.

(35) How many eggs can people eat, depending on their age?

[5] Answer (35a) was found following question (35) as a section title on a French website: https://oeuf-info.fr/infos-nutrition/

a. Up to the age of 10: one egg a day. From age 10 upwards: children and teenagers can eat as many eggs as adults if they wish.

(36) Can it really make a difference if you plant based on the moon cycles? Will your grass grow faster or slower depending on which "phase" you mow it in? [`enTenTen21`]

Dependence questions can be embedded. In (37), although the SELON *A* phrase syntactically modifies the whole main clause, it is semantically dependent on the embedded interrogative *B*. As such, the whole sentence is not a question, but an attitude ascription about a dependence question, as paraphrased in (37a).

(37)

en fonction de	[$_A$ combien	il	te	reste	d'	éléments dans
SELON	how_many	EXPL	you.DAT	remain	of	elements in

ton	objet]	tu	peux	savoir	[$_B$ depuis	combien	de	temps	il	est
your	object	you	can	know	since	how_much	of	time	it	is

en train de	se	dégrader]
PROG	REFL	degrade

'Depending on [$_A$ how many (radioactive) elements you have left in your object], you can find out [$_B$ how long it has been decaying].'
[`CIENSFO`: 201]

a. You can know what the dependence relation f is between the element count and the decay duration.

Theiler et al. 2019 cannot model inquisitive ADUs because their definition produces assertive issues.

6.2.3 Presuppositions of dependence questions

In Theiler et al.'s definition (12), both conjunction of conditionals and non-triviality are at-issue. According to Roelofsen (2015), any content that should normally arise at-issue in the derivation of a question is actually rendered as a presupposition. This rule correctly predicts that the exam question (35) presupposes that the answer is a non-trivial dependence function.

However, the non-triviality presupposition seems to be absent from canonical questions. For example, (38) suggests that the repercussions will likely be different for different outcomes. However, the interlocutor may answer and negate non-triviality without background disagreement, e.g. with an unconditional statement (38a). Therefore, non-triviality simply appears to be an implicature in canonical dependence questions. However, it is unclear why this weakening occurs (only in this case).

(38) Depending on the outcome, what will be the regional and national repercussions of this initiative? [`enTenTen21`]

   a. Regardless of the outcome, the repercussions will be the same: an increase in prices.

7 Towards a uniform definition of dependence

Here, we provide a general definition for dependence that abstracts from any model. Based on Theiler et al.'s lexical entry, the core ingredients are: alternatives in the antecedent and consequent, a functional correspondence f between them, a modal base, a conjunction of conditionals restricted to the modal base, and a non-triviality condition on the correspondence.

The problems with their definitions are: (i) consequent alternatives may not always have truth conditions, and (ii) the output relation is not always the assertion of the existence of f. In particular, the dependence question (35) must be able to quantify over the set of dependence functions f, and the low-scoping example (31) must recover the true correspondence f as a selection criterion. To capture these behaviors, we assume that a dependence relation provides a dependence function variable f by default. An independent (covert) operator should be in charge of existentially closing that variable in adequate contexts.

Given two semantic arguments A (the antecedent) and B (the consequent) and a modal base M, we define the relation "B depends$_M$ on A" as the two components:

1. requires: A raises alternatives $\textsc{alt}(A)$ and B raises alternatives $\textsc{alt}(B)$
2. provides: a function $f : \textsc{alt}(A) \rightarrow \textsc{alt}(B)$, such that
   a) for all world $w \in M$ and alternative $A_i \in \textsc{alt}(A)$, the evaluation of $f(A_i)$ at w is conditionalized by the evaluation of A_i at w,
   b) and f is non-trivial (i.e. 12b)

For assertions, evaluation is truth; for commands, evaluation is satisfaction; and for questions, evaluation is resolution. If not specified, the modal base is computed as $M = \bigcup \textsc{alt}(B)$.

8 A uniform model for different utterance types

8.1 Background on dynamic inquisitive semantics

8.1.1 Motivations

To formalize the definition of dependence given in §7, we need a semantics capable of modeling semantic objects interpreted from different clause types, and a rich enough compositional theory to model the different dependence levels observed above: speech-act level, clause level, and predicate level.

The left periphery proposed by Rizzi (1997) (and subsequent works) offers a substantial basis for a theory of adjuncts. We aim to define a single lexical entry for SELON and to explain the different readings in terms of different attachment sites on the spine of logical form.

Inquisitive semantics models assertions and questions in an integrated way. Thanks to this property, lexical entries for predicates allowing both assertions and questions as arguments can be defined uniformly. Therefore, we base our model on dynamic inquisitive semantics (Roelofsen & Dotlačil 2023). This framework extends inquisitive semantics with discourse referents and the compositional theory based on Rizzi's left periphery. Although our analysis is unrelated to anaphora, discourse referents are used as a way to co-index variables.

8.1.2 Logical operators

In dynamic inquisitive semantics, clauses are interpreted as update functions $\mathcal{V}$ between contexts. A context c is a downward-closed set of information states. An information state s is a set of pairs $\langle$world w, assignment function $g\rangle$. Contexts extend the notion of issues from inquisitive semantics. Alternatives of contexts $\textsc{alt}(c)$ are defined similarly. Update functions can raise inquisitiveness, i.e., generate an inquisitive output context from an assertive input context.[6]

Various operators control inquisitiveness. The inquisitiveness-elimination operator "!" prevents an update function $\mathcal{V}$ from raising inquisitiveness. In particular, if c is an assertive context, $\textsc{alt}((!\mathcal{V})(c))$ contains a single alternative, namely $\bigcup \mathcal{V}(c)$. For example, if the alternatives of $\mathcal{V}(c)$ are "d sleeps" for every individual d, then $\textsc{alt}((!\mathcal{V})(c)) = \{$"someone sleeps"$\}$. Thus, "!" acts like an existential closure operator for alternatives.

[6] We refer to Richard (2026: §3) for a short overview, formal definitions, and illustrations of dynamic inquisitive semantics.

On the contrary, the conditional inquisitiveness closure operator $\langle ? \rangle$ ensures that an update function raises inquisitiveness. If $\mathcal{V}$ already raises inquisitiveness, $\langle ? \rangle$ is vacuous. Otherwise, $\langle ? \rangle\mathcal{V}$ raises the polar question "$\mathcal{V}$ or not $\mathcal{V}$".

Noun phrases, including wh-phrases, bear a discourse referent. Given a discourse referent u, the logical operator $?u$ is an update function that raises as many alternatives as there are individuals d that u can be valued with.

8.1.3 Overview of the left periphery

Now, let us describe the left periphery and the contribution of each head, listed from the highest to the lowest.

Following Speas & Tenny (2003), we assume a speech-act phrase (SAP). It contributes information about the speech act and the participants: $\mathsf{assert}_{S,H}$, $\mathsf{command}_{S,H}$ or $\mathsf{question}_{S,H}$. Existential closure of imperative ADUs can be explained by a pragmatic principal, namely global discharge (Kaufmann 2016: §5.3).

The complementizer phrase (CP) encodes the syntactic clause type. The head of declarative clauses is French *que*, English *that* or covert DEC otherwise. They eliminate inquisitiveness raised below in the derivation tree. The head of interrogative clauses is French *est-ce que* or *si*, English *whether* or covert INT by default. They ensure that the interpretation is inquisitive.

The focus phrase (FocP) is the location of focus realization. Focused elements, including wh-words, move to the specifier of FocP to co-index the covert head FOC with their discourse referent. Spec FocP is the surface position of ex situ wh-words. If co-indexed, the head FOC_u raises u-domain alternatives. Before interpretation, focused elements are reconstructed through an across-the-board movement.

Finally, FinP is the finite phrase and VP the verb phrase. The contribution of the covert heads below SAP is given in equation (1.39). The overt realizations have the same respective entries. The semi-colon ";" is the conjunction of update functions.

$$\begin{aligned} [\![\text{DEC}]\!] &= \lambda\mathcal{V}.\,!\mathcal{V} & \qquad [\![\text{FOC}]\!] &= \lambda\mathcal{V}.\,!\mathcal{V} \\ [\![\text{INT}]\!] &= \lambda\mathcal{V}.\,\langle ? \rangle\mathcal{V} & \qquad [\![\text{FOC}_u]\!] &= \lambda\mathcal{V}.\,!\mathcal{V};?u \end{aligned} \tag{1.39}$$

The complement of SELON is a CP, so that it can be an interrogative CP, a disjunction of declarative CPs, or contain only an NP, which is coerced as a concealed question by moving it to Spec FocP to co-index FOC.

To model the different dependence levels, we assume that the SELON A adjunct can attach at various places on the spine of the logical form. Figure 2 displays the possible attachment sites for the dependence levels described above. Attached to

SAP, ADUs transform a disjunction of speech acts into a conjunction of conditionalized speech acts. Attached to VP, ADUs provide an optional argument of the predicate, and their consequent is passed by the predicate during derivation.

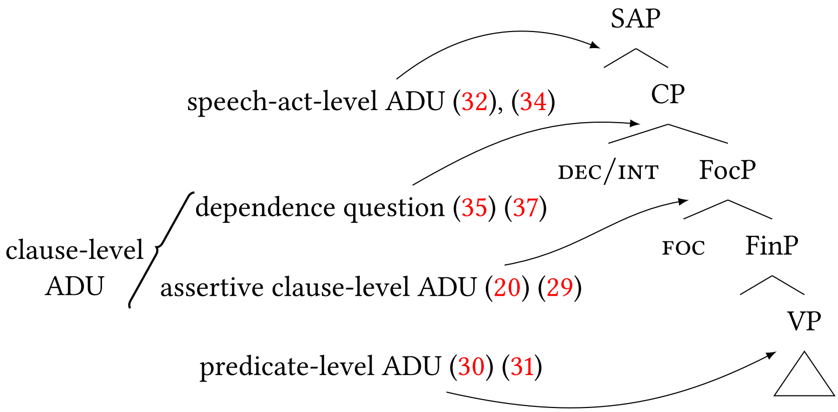

Figure 2: Attachment sites of the SELON A adjunct on the logical form for different adjunct dependence utterances

8.2 A single lexical entry

Now, let us focus on clause-level ADUs, i.e. dependence questions and assertive clause-level ADUs. We ignore the speech-act phrase here.

As explained in §7, we need to express that, by default, a dependence relation *provides* a dependence function instead of asserting its existential closure. Richard (2024b) describes a recipe to turn a modal operator into an inquisitive version that propagates inquisitiveness, and so letting its output be inquisitive. This change consists of inverting the universal quantifier over information states ($\forall w \in s$) and an existential quantifier.

Viewing SELON A as a complex modality, we follow Richard's idea by pulling the $\exists f$ quantification. We interpret SELON in dynamic inquisitive semantics as (40), using (41) and leaving fmd unchanged (12).

(40) $[\![\text{SELON}]\!] := \lambda \mathcal{V}, \mathcal{W}.\ \lambda c, s.\ s \in c \wedge \mathsf{dep}_M(\mathcal{V}(c), \mathcal{W}(c))(s)$

(41) $\mathsf{dep}_M(A, B) := \lambda s.\ \exists f : \text{ALT}(A) \rightarrow \text{ALT}(B).\ \forall w \in s.\ \mathsf{fmd}(f, M)$

Definition (40) indicates that $[\![\text{SELON}]\!](\mathcal{V}, \mathcal{W})$ is an update function that takes an input context c and outputs the context composed of the information states s

that were already in the input and that satisfy a dependence condition between $\mathcal{V}$ and $\mathcal{W}$ evaluated at c. Definition (41) indicates that $\mathsf{dep}_M(A, B)$ is a context composed of the information states s that agree on a function f satisfying function modal dependence (fmd). By putting $\exists f$ first, this creates as many alternatives as functions f satisfying $\mathsf{fmd}(f, M)$. Therefore, $\mathsf{dep}_M(A, B)$ could be paraphrased as: What is the dependence function f between A and B?

In clause-level ADUs, SELON A must attach above FocP, where the focused elements move. In dependence questions like (35), SELON A attaches at CP, above INT, so that the consequent B can be an interrogative CP. However, in assertive main-clause ADUs, SELON A needs to attach below DEC so that the inquisitiveness-elimination operator "!" provided by DEC existentially closes the f alternatives. In other words, $!\mathsf{dep}_M(A, B)$ is equivalent to Theiler et al.'s definition (if we ignore the assignment functions). As a consequence, the lexical entry (40) captures both dependence questions and assertive clause-level ADUs thanks to the properties of the declarative CP head.

9 Conclusion

In many languages, dependence can be expressed with a finite verb, like English *depend*, but also with an adjunct clause. We call adjunct dependence utterance (ADU) a construction consisting of a main clause modified by a phrase expressing dependence. Adjunct dependence utterances are cross-linguistically diverse. In English, we have the verb phrase *depending on A*, in French the prepositional phrase *selon / suivant / en fonction de A* and in German the complementizer phrase *je nachdem A*. They all share a similar syntactic structure, and we use SELON to denote any dependence-introducing adjunct.

ADUs SELON A, B are complex conditional constructions. We call antecedent A the complement of SELON and consequent B its second argument, often the main clause. ADUs accept various syntactic categories as their antecedents: interrogative clauses, noun phrases as concealed questions, and declarative CP disjunctions (only in French). The consequents of assertive ADUs are declarative clauses. Clause-level ADUs quantify over this entire consequent clause. However, there exist predicate-level ADUs, typically in combination with *not be the same* or *choose*. We also observe imperative ADUs, which give a command that should be selected from a list based on a parameter. Finally, we illustrated different kinds of inquisitive ADUs: speech-act questions and questions about dependence.

This article captures this variation by providing a general definition of dependence. A dependence relation requires that the antecedent and consequent raise alternatives. It provides a non-trivial dependence function f, i.e. a correspondence between the antecedent alternatives A_i and consequent alternatives, contributing a conjunction of the expressions: $f(A_i)$ conditionalized by A_i.

An adjunct SELON A is a modal quantificational adverbial operator. It uses a modal base, derived by default as the conjunction of the consequent alternatives. Consequent alternatives can be raised by focus.

We showed that the different dependence levels can be explained by different attachment sites of the SELON A adjunct on the logical form spine. We provided a uniform model for clause-level assertive and inquisitive ADUs by defining a single lexical entry for SELON.

Abbreviations

COND	conditional tense
DAT	dative case
EXPL	expletive pronoun
FUT	future tense
GEN	genitive case
NEG	negative clitic
PRO	pronominal
PROG	progressive aspect
REFL	reflexive pronoun
SJV	subjunctive mood

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